

1-A:

A – The cost of producing the last extra unit represents the ‘marginal cost’. The value the consumers place on the product represents the ‘price’.

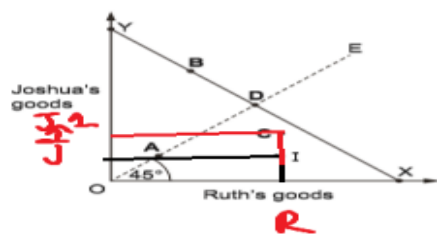
When marginal cost is equal to the price, this results in allocative efficiency. This is why, option A becomes correct.

B – When the cost of producing the last extra unit which represents the ‘marginal cost’ reduces, economic efficiency is not achieved.

C – When the total cost of production is equal to the value consumers place on the total output, this does not represent economic efficiency either.

D – When the total revenue is at its maximum, this is when the marginal revenue is zero. And, this does not result in efficiency either.

2-C:



→ The Pareto criterion states that a resource allocation or an economic situation is considered Pareto efficient if it is impossible to make any individual better off without making someone else worse off. In other words, a “Pareto improvement” occurs when a change in the allocation of resources results in at least one person being better off, without making anyone else worse off.

A – At point “A”, the output of Joshua is the same, whereas the output of Ruth is falling. This is why, Pareto improvement is not occurring here.

B – At point “B”, the output of Joshua is increasing and the output of Ruth is decreasing. This is why, Pareto improvement is not occurring here.

C – At point “C”, the output of Joshua is increasing and the output of Ruth remains constant. “Pareto improvement” occurs when a change in the allocation of resources results in at least one person being better off, without making anyone else worse off.

This is why, key C, is correct.

D – At point “D”, the output of Joshua is increasing but the output of Ruth is falling, which represents that Pareto improvement will not be occurring.

3-D:

- A- Calculation of costs is easier to compute in **private sector investment projects** since they only consider **private costs**, which have **monetary values attached** to them.
- B- This is incorrect, as all projects, **be it a public sector or private sector project**, involve risk. This cannot be said that **private sector projects involve less risk**.
- C- Every project, **be it a public sector or private sector project**, involve benefits which will occur in future.
- D- This is correct, as since public sector projects take **externalities(external costs and benefits)** in consideration, which **do not go through the market**, and therefore **do not have monetary values attached to them**, there is a **difficulty in estimation**, which results in **uncertainty**. **Shadow prices** are hypothetical or imputed prices that represent the value of an externality or an unpriced factor in a market or economic activity. **For instance, for a transportation project, analysts might estimate the value , through shadow prices, of reduced travel time for the third party (positive externality) or the value of air pollution-related health costs (negative externality) to be reflected in the cost-benefit analysis.**

This is why , option D, becomes correct.

4-A:

A - An individual demand curve slopes downwards to the right because of the working of the Law of Diminishing Marginal utility.

As an individual starts consuming a good one after the another, the satisfaction obtained upon consuming an additional unit or the marginal utility falls, and thus the willingness to pay a higher price for that particular product falls. This is why option A is correct.

B- Consumers being faced with choices of competing products represents the concept of "substitutes"

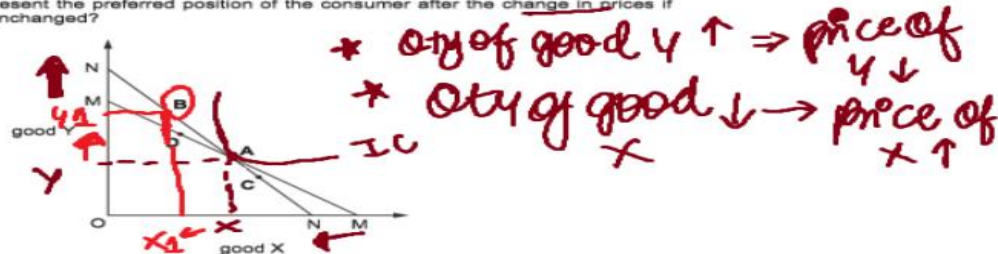
C – Seller's being willing to accept lower prices represents producer's perspective, and we are interested in consumer's perspective.

D – The fall in average costs as a result of a rise in the scale of production represents "economies of scale"

5-B:

The diagram shows budget lines of a consumer choosing between two goods, X and Y. Initially the budget line is MM and the consumer's preferred position is at point A. Subsequently the money prices of both goods change shifting the budget line to NN.

Which point could represent the preferred position of the consumer after the change in prices if their tastes remained unchanged?



A- Option A is eliminated since it shows the initial consumer equilibrium.

B- Option B is correct since it is on the new budget line “NN”, and on it, the qty of Y is shown to increase and the qty of X is shown to decrease.

C- Option C is incorrect since here the qty of Y is shown to decrease and the qty of X is shown to extend.

D- Option D is eliminated since it is on the old budget line “MM”

6-D:

In a perfectly competitive market:

- A single firm's actions do not affect the market price (**because the firm is a price-taker**)
- The firm can sell all it wants at the prevailing market price. **This represents that the demand is perfectly elastic.**
- The market price always clears the market (supply equals demand)

Since the firm's actions do not affect the market price, it implies that the firm is a small part of the overall market, and there are 1000s of producers producing the homogenous product in the entire market. This is a characteristic of perfect competition. This explains why key D will be correct.

A. Imperfect competition refers to markets that do not meet the criteria for perfect competition.

Some features of imperfect competition are as follows.

1. **Non-price competition:** Firms compete through advertising, product differentiation, and other non-price factors, rather than just price.
2. **Barriers to entry:** New firms face obstacles entering the market, such as high startup costs, patents, or government regulations.
3. **Product differentiation:** Firms produce differentiated products, which can lead to monopoly power and price-making ability.
4. **Externalities:** The production or consumption of a good affects third parties, such as pollution or positive externalities like technology spillovers.

5. Information Asymmetry

- Definition: Firms or consumers have unequal access to information
- Effect: Leads to market failures
- Example: A company knows more about the quality of their product than potential customers, or a buyer has more information about their reservation price than the seller.

6. Monopsony Power

- Definition: A single buyer dominates the market
- Effect: Allows them to influence prices
- Example: A large retailer is the only buyer of a particular product, giving them significant bargaining power over the suppliers.

B. Monopoly refers to a market structure in which a single firm supplies most of the market with a particular good or service, giving it significant market power.

C. Oligopoly refers to a market structure in which a small number of large firms (4-5) compete with each other, often leading to interdependent decision-making and strategic behavior.

7-C:

A. Barriers to entry can actually facilitate cooperative behavior among existing firms. With barriers in place, firms may feel more secure in their market position and more likely to engage in collusion to maximize their joint profits.

B. If firms can easily observe each other's prices, they can more easily coordinate their pricing strategies and maintain a collusive agreement.

D. Stability in demand and costs can provide a favorable environment for cooperative behavior. When market conditions are stable, firms may find it easier to maintain a collusive agreement.

C. Product differentiation, on the other hand, can lead to non-cooperative behavior. When firms have unique products, they may focus on their individual brand and market share, making it more difficult to maintain a collusive agreement. This is why, **only firms in a non-collusive oligopoly employ product differentiation, since they engage in non-price competition.**

8-B:

A – When there are a large number of firms in the industry, this represents a greater degree of competition, and now firms will compete with each other and make attempts to be efficient, in order to be able to cut their costs so that they can stretch the difference between revenues and costs, and in turn achieve the objective of “profit-maximization”

B – When there are a large number of shareholders and a few paid managers, this represents a large public limited company and it is most likely that this will give existence to the principal-agent problem. Divorce of ownership and control will exist because managers who are in control may want to achieve the objective of sales-revenue maximization, since their salaries and bonuses are based on revenues,

and not profits, as compared to shareholders, who aim for “profit-maximization” to be able to receive a higher payout in form of dividends. This is why key B is correct.

C – When there is a large investment expenditure required, this represents that the firm operates with the objective of “profit-maximization”, since there will be a requirement of substantial retained profits to finance re-investment in the business.

D – Again, when there is a large output being produced, this will lead to cost advantages in form of economies of scale, and when there is a fall in Long-Run Average costs, this will be a reason why “profit-maximization” is achieved.

9-C:

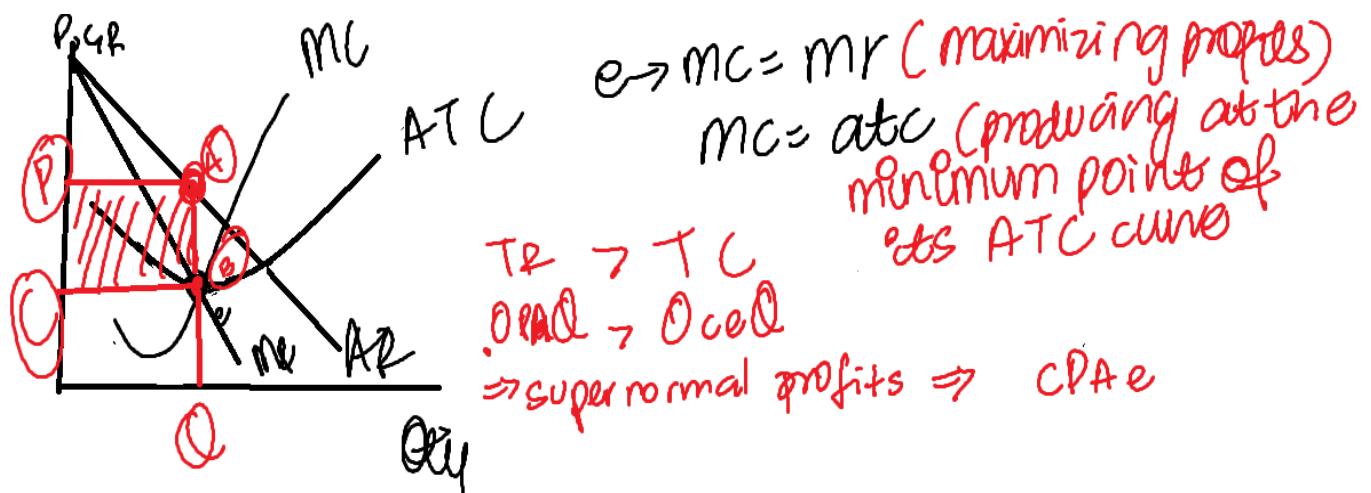
A – Since it is stated in the question that the firm produces at the minimum point on its Average total cost curve, it can be said that, marginal cost will be equal to Average total cost. This is because, when the marginal cost curve intersects with the Average cost curve, Average total cost is minimized.

B – The question also states that the firm is maximizing profits which means that it can be said that Marginal cost is equal to Marginal revenue.

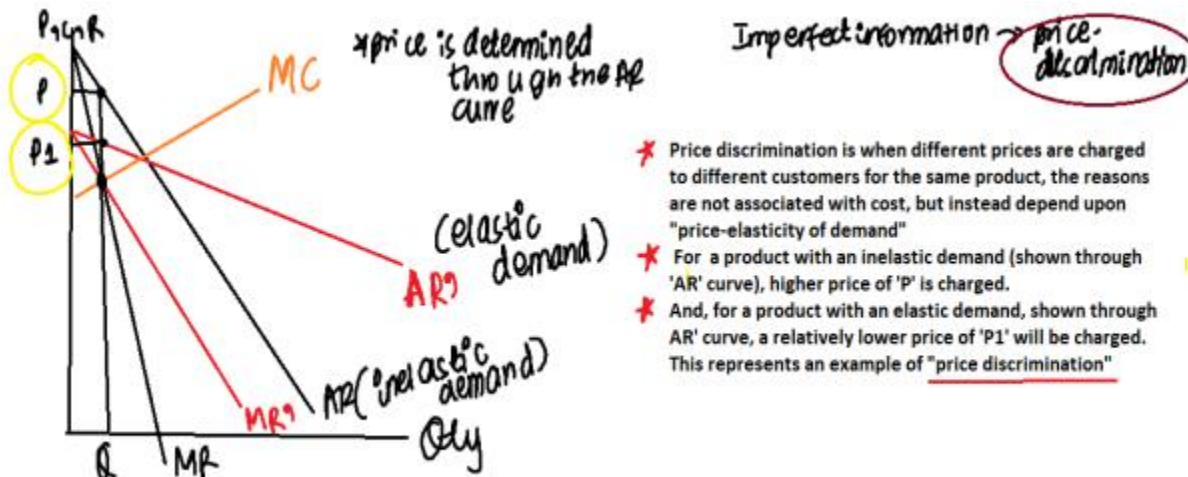
Why C and not D:-

Since the firm is operating where $MC=MR$, and is also producing at the minimum point on its ATC curve, this represents that Average revenue will be greater than Average total costs. When because of this, total revenue exceeds total cost, there will be super-normal profits being achieved.

However, this means that normal profits that are achieved when $AR=AC$ will not be made, and hence, key C becomes the right option.



10-B:



A – Even if there are many buyers, due to **availability of information**, price discrimination cannot be done.

B – It can be seen in the diagram above that , **there must be markets with "different price-elasticities of demand"**, in order for price discrimination to be done.

C – Price discrimination is mainly "done for the **same product**", and therefore, product differentiation can make price discrimination weaker.

D – Even if markets are separate, due to **availability of information**, price discrimination cannot be done. **For, price discrimination to be effective, there must be "imperfect information"**

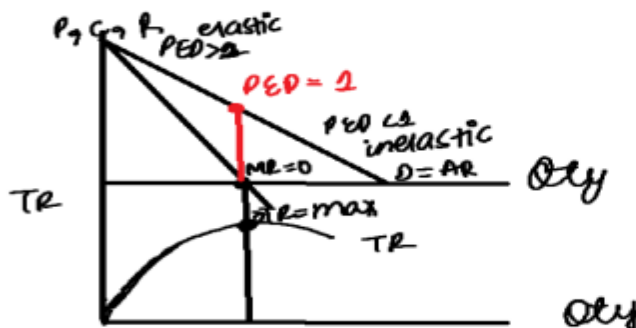
11-C:

A – This is incorrect as a downward sloping demand curve (AR curve) represents the existence of imperfect competition, and here, average revenue is not equal to marginal revenue.

And, it is only in case of perfect competition that the Average revenue is equal to marginal revenue.

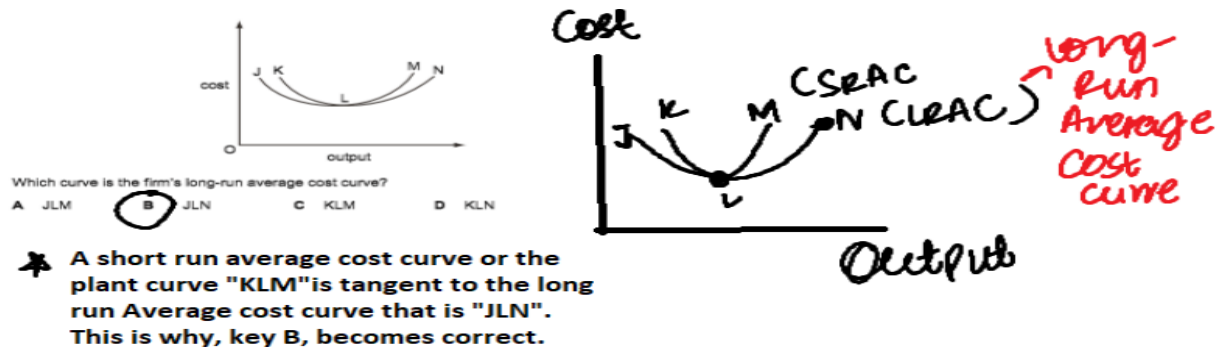
B – Average revenue is zero, when the demand curve (AR curve) touches the x-axis, which represents the point where $PED=0$ (perfectly inelastic)

C – In-order to maximize Total Revenue, the marginal revenue must be "0", and this is shown in the diagram below. This is why, key C, will be correct



D – Again , as mentioned above in option A , average revenue and marginal revenue can never be equal under “imperfect competition”

12-B:



13-B:

The employment size distribution in Country X shows a higher percentage of small firms (<50 employees) and a lower percentage of large firms (>250 employees) compared to Country Y. This suggests that Country X has a more entrepreneurial economy, with more small and medium-sized enterprises. This is likely due to a larger supply of entrepreneurs in Country X, who are more likely to start and run their own businesses.

This explains why key B will be correct.

A. A bigger domestic market in Country X would likely lead to more large firms, not more small ones.

C. Greater involvement in international trade would likely lead to more large firms in Country X, as they would need to be competitive in global markets.

D. We only have data on employment size distribution of manufacturing firms in two European countries, and based on this, we cannot conclude if country X makes use of advanced technology.

14-C:

Since the firm faces a downward sloping demand curve, it can be said that we are operating in “imperfect competition”

A – The nominal(current) price of output, and the nominal wage rate of labor would not help to derive a downward sloping demand curve.

B – The marginal cost of labor curve does not help to derive the demand curve for labor, because Marginal cost of labor is only equal to the Marginal revenue productivity at the profit-maximizing level of employment.

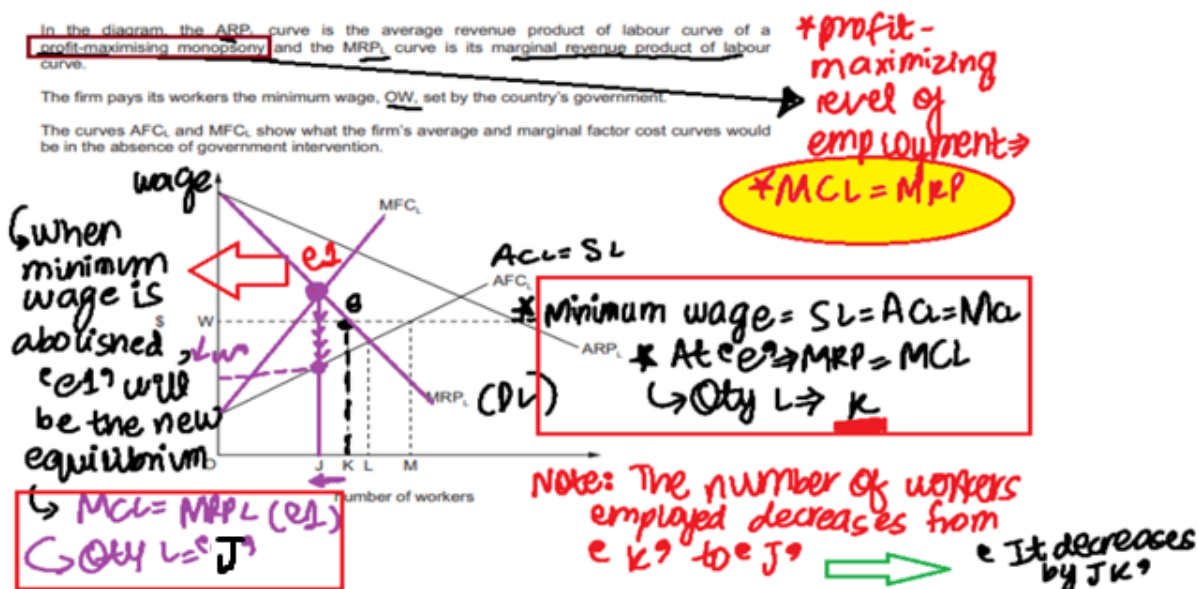
C - The **MRP of labor** refers to the **additional revenue generated by hiring one more unit of labor**. It represents the **contribution of that worker to the firm's total revenue**. As a firm hires more workers, the **MRP reflects the extra output produced and the associated revenue**. And the **“Marginal revenue productivity curve”** is equal to the **“demand for labor curve”**, because the firm takes hiring decisions on the basis of **“the contribution of each worker to the output” and to the “revenue”**.

And, since MRP is obtained by multiplying “MPP” with “MR”, it can be said that the Marginal Revenue productivity curve is obtained from the **schedules of Marginal physical product of labor, and of the marginal revenue**. This is why, key C, becomes correct.

D – The schedules of the marginal physical product of labor and the money wage rate do not help to obtain the Marginal Revenue productivity curve.

This explains why key C will be correct.

15-A:



16-D:

The removal of a sales tax on good X will bring about a decrease in the costs of production, and subsequently, there will be an increase in supply of good X. This will subsequently increase the demand for labor, which is a derived demand, as firms will require more labor to increase production. This explains why key D will be correct.

- A. A decline in marginal physical productivity would make workers less efficient, leading to reduced employment as firms will reduce the demand for labor because of this.
- B. A tariff on a complementary good would make it more expensive, leading to reduced demand for good X and subsequently there will be a reduction in employment.
- C. An employer's contribution to a state welfare scheme would increase labor costs, making firms less likely to hire workers.

17-B:

Top rates of income tax: Increase

Increasing top rates of income tax will make the tax system more progressive. This is because higher-income individuals will be paying a larger share of their income in taxes, reducing the after-tax income inequality.

Indirect taxes: Reduce

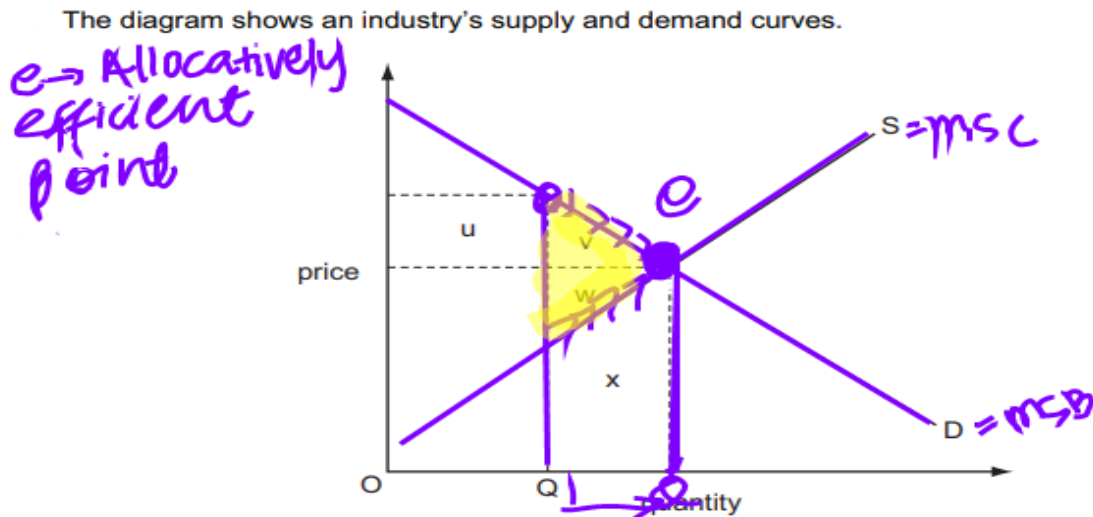
Reducing indirect taxes will make the tax system less regressive. Indirect taxes, such as sales taxes or VAT, are regressive because they disproportionately affect low- and middle-income households, who spend a larger portion of their income on consumption. By reducing indirect taxes, the tax system will become less regressive.

Value of state benefits: Increase

Increasing the value of state benefits (means tested benefits) will provide more support to low-income households, thereby reducing poverty and income inequality. This will help to reduce the income gap between the rich and the poor, making the income distribution more equal.

This explains why key B will be correct.

18-C:



Since output Q_1 represents an allocatively inefficient level of output, this implies that there would have been a deadweight loss of $v+w$ when the output was restricted to this level.

But when this restriction is removed and output is increased to Q again, there will be a net gain in economic welfare of $v+w$.

This explains why key C is correct.

19-A:

The correct answer is A: benefits paid to the unemployed.

National income measures the total income earned by a country's citizens, including:

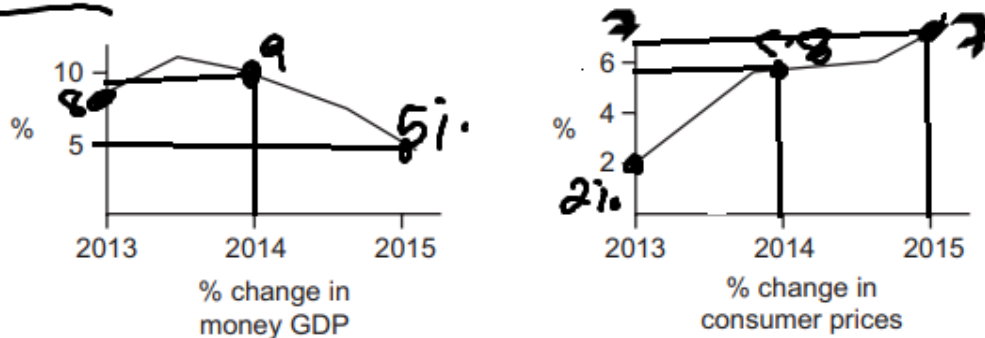
- Dividends paid to shareholders (B)
- Profits retained by companies (C)
- Rent paid to landlords (D)

However, benefits paid to the unemployed (A) are not included in the measurement of national income, as they are considered transfer payments, not earned income.

Transfer payments are payments made by the government to individuals without any corresponding production or contribution to GDP. Other examples of transfer payments include pensions, social security benefits, and welfare payments.

20-A:

The graphs show percentage changes in money GDP and consumer prices in a country between 2013 and 2015.



A) This option is correct because the data shows that consumer prices rose by 2% in 2013, 5.8% in 2014, and 7% in 2015, which means they continued to rise throughout the period. Additionally, money GDP rose by 8% in 2013, 9% in 2014, and 5% in 2015, which means it also continued to rise throughout the period.

B) This option is incorrect because while money GDP continued to rise, the growth rate slowed down in 2015 (5%) which is lower than the growth rate of consumer prices (7%). This means that in real terms, GDP may have actually declined in 2015. Therefore, we cannot conclude that real GDP grew throughout the period.

C) This option is incorrect because the data shows that money GDP rose by 9% in 2014 and by 5% in 2015. While the growth rate slowed down, money GDP did not fall. Therefore, we cannot conclude that money GDP fell more rapidly between 2014-2015 than between 2013-2014.

D) This option is incorrect because there is no information about money incomes falling.

21-B:

Most developing economies typically have:

- A high ratio of government debt to GDP, due to high expenditure needs which cannot be met by tax revenues which are low due to a lower tax base, and often there is a reliance on external borrowing.
- A low average propensity to save, as households and businesses in developing economies often have limited income and resources, leading to lower savings rates.

22-C:

The movement of labor from developing countries to developed countries can have the following effects:

Economic growth: Increase

- Immigrant workers contribute to the labour force, increasing the productive capacity of the economy. In other words, since the quantity of labor (i.e. a f.o.p rises), the long-run Aggregate supply of the economy will increase. This indicates potential growth.

.Pressure on wage rises: Decrease

- An influx of immigrant workers can lead to a surplus of labor in the economy, which will result in a downward pressure on wage rises. This explains why C becomes correct.

23-A:

Cyclical unemployment is associated with a deficiency in aggregate demand, which occurs during economic downturns or recessions. This type of unemployment arises when there is not enough demand for goods and services, leading to a decrease in production and employment.

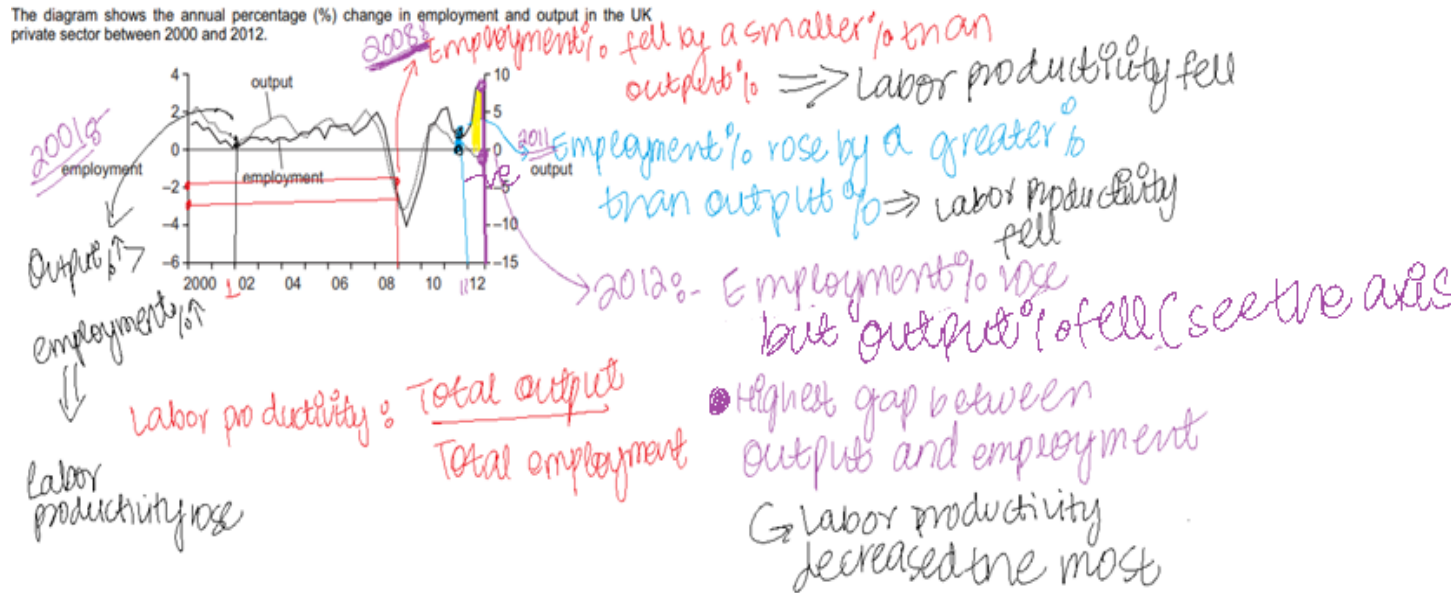
B) Frictional unemployment occurs when workers are temporarily unemployed in between jobs. This can arise because of the process of searching for a new job, and it is not directly related to aggregate demand.

C) This type of unemployment occurs when there is a mismatch between the skills and qualifications of workers and the requirements of available jobs. This occurs due to changes in the structure of the economy, or in other words, because of changes in the pattern of demand and supply in the economy.

D) Voluntary unemployment occurs when workers choose to voluntarily not work at the going wage rate.

24-D

The diagram shows the annual percentage (%) change in employment and output in the UK private sector between 2000 and 2012.



25-A:

A) With more ATMs available, people are more likely to keep their money in bank deposits and withdraw cash as needed, rather than keeping large amounts of cash on hand. This is because, they know they can withdraw cash whenever needed. **This explains why key A will be correct.**

B) A reduction in interest rates on bank deposits would likely lead to a decrease in the public's willingness to deposit in bank since return on savings will fall, potentially increasing the ratio of cash to bank deposits, since people are more likely to hold cash.

C) The closure of bank branches would likely make it more difficult for people to access their money, potentially leading to an increase in the ratio of cash to bank deposits. This is because, they are more likely to hold money for emergencies.

D) The introduction of annual charges for the use of credit cards would likely lead to a decrease in the use of credit cards. When people do not buy on credit, they will be more likely to hold cash for transactionary purposes, which will increase the cash to deposit ratio.

26-C:

In an economy with no government sector

gx $\begin{cases} C = \text{consumption} \\ I = \text{investment} \\ X = \text{exports} \\ M = \text{imports} \\ Y = \text{national income.} \end{cases}$

If $C = 20 + 0.75Y$, $I = 60$, $X = 120$ and $M = 120$, what will be the equilibrium level of Y ?

A 80 B 280 C 320 D 400

① $Y = A E$ (macroeconomic equilibrium)

$$Y = 20 + 0.75Y + 60 + (120 - 120)$$

$$Y = 20 + 0.75Y + 60$$

$$Y = 80 + 0.75Y$$

$$Y - 0.75Y = 80 \Rightarrow 0.25Y = 80$$

$$\rightarrow Y \rightarrow 80 / 0.25$$

$$\rightarrow Y \rightarrow 320$$

27-D:

There are net leakages when the answer of **leakages-injections** is positive, or in other words, leakages are greater than injections.

Private sector surplus: $S > I$: Yes

When savings (leakage) are greater than investment expenditure (injection), this represents net leakages into the circular flow.

Public sector: $T > G$: Yes

When taxes (leakage) are greater than government expenditure (injection), this represents net leakages into the circular flow.

Trade sector: $X > M$: No

For there to be net leakages into the circular flow, export revenue that is an injection into the circular flow should be less than import expenditure which is a leakage from the circular flow.

28-B:

The Philips curve represents a historical inverse relationship between inflation and unemployment.

The table gives an economy's unemployment rate and inflation rate for a five-year period.

year	unemployment rate %	inflation rate %
2010	7.6	2.6
2011	6.2	1.6
2012	5.8	1.8
2013	5.9	2.0
2014	5.8	2.0

Handwritten annotations: A purple bracket on the left side of the table spans from 2010 to 2014. Purple arrows point down from 7.6 to 6.2, and from 6.2 to 5.8. A pink arrow points up from 5.8 to 5.9. A blue arrow points up from 1.6 to 1.8. A cyan arrow points right from 2.0 to 2.0. To the right of the table, the words 'tve', '-ve', and '+ve' are written in purple, pink, and blue respectively, corresponding to the changes in inflation.

There is only a negative relationship between these variables in between 2011-2012, which explains why key B becomes correct.

29-C:

A) Since there are already low incomes earned in the poor country, increasing the savings ratio will exacerbate short-term actual economic growth. Additionally, it will not even remedy this productivity gap.

B) Allowing workers from poor countries to migrate to rich countries may lead to a brain drain in poor countries, as the most skilled workers leave in search of better opportunities. This could exacerbate the productivity gap, rather than remedy it.

C) Investing in education in poor countries can help to improve workers' skills and knowledge, enhancing their ability to use technology effectively and increasing their productivity. This is the most effective way to remedy the productivity gap.

Note: Education can help to build human capital, which is essential for economic growth and development. **This represents an interventionist supply side policy tool.**

D) While removing trade barriers can improve market access for poor countries, it may not address the productivity gap. The productivity gap is often due to a lack of education and skills, rather than a lack of market access.

This explains why key C is correct.

30-A.

A rise in interest rates in a country makes it more attractive to foreign investors, who can earn higher returns on their short-term investments. This leads to an increase in the inflow of short-term foreign capital (**namely hot money inflows**) , as investors seek to take advantage of the higher interest rates.

B) A rise in interest rates can actually decrease company profits, as higher interest expense can reduce the net profits earned.

C) Higher interest rates can lead to a decrease in private investment, as higher borrowing costs can make investment projects less attractive.

D) Higher interest rates can lead to higher borrowing costs which can reduce company profits and make shares less attractive to investors, and when demand for shares fall, share prices are likely to decrease.

